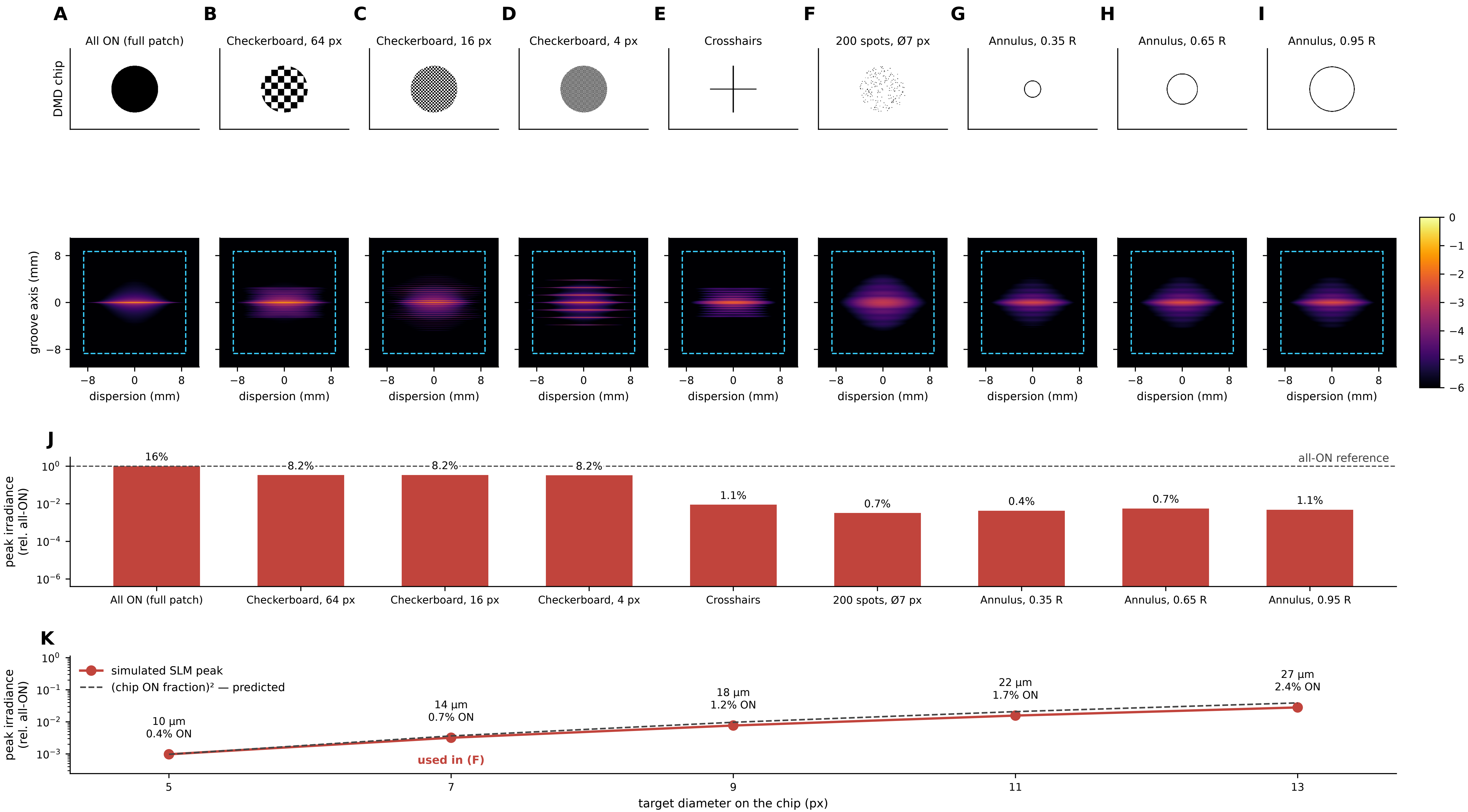

Irradiance at the SLM is the Fourier transform of the DMD pattern, smeared by the laser spectrum. Scalar Fourier propagation of the merged arm to the Meadowlark HSP1K, at the bench-confirmed f7 = 300 mm of handoff rev 5. Every optical constant, including the lens prescription and grating angles, is parsed live from docs/optics_handoff.md and cross-checked against it. **(A–I)** Top row, the binary frame written on the DMD, confined to the Ø5.0 mm illuminated disc; bottom row, the resulting time-averaged irradiance on the liquid crystal, shown as $\log_{10}$ relative to the all-ON peak over a common 6-decade scale and identical ±11 mm axes, so panels are directly comparable. Dashed cyan square is the 17.4 mm LC aperture. The horizontal axis is the grating's dispersion direction; because the chip is clocked 45°, each transform is rotated into this lab frame, which is why chip-aligned features (D, E) appear on the diagonals. **(A)** A filled patch puts everything into a single DC order, which the grating then draws out into a spectral line of 0.01 × 7.1 mm ($1/e^2$, groove × dispersion). That ~400:1 smear is what keeps the LC below its damage limit, and why the air-breakdown interlock sits upstream of the grating instead. **(B–D)** Checkerboards at 64, 16 and 4 px place first orders at a radius that grows as 1/period, walking light outward from DC. **(E)** Crosshairs transform to a cross rotated 90° from the bars that produced it. **(F)** 200 targets of Ø7 px (≈14 µm at the sample), the realistic experiment: a broad speckled envelope with no concentration. **(G–I)** Annuli at 0.35, 0.65 and 0.95 of the patch radius give Bessel-ring transforms whose fringe spacing narrows as the ring grows. **(J)** Peak irradiance per pattern relative to the all-ON frame (log axis, dashed line = that reference); the percentage above each bar is the fraction of the whole chip switched ON, which is the quantity the 50% load-time cap tests. **(K)** Peak irradiance for 200 targets as target DIAMETER is swept 5–13 px (≈10–27 µm), spanning 29× — bigger targets both carry more power and transform to a narrower pupil distribution. The dashed line is the (ON fraction)² law the DC order obeys, drawn as a prediction anchored on the smallest spot rather than fitted; the agreement is why target size, not target count, is the strong lever on LC irradiance. Method: field = pattern × Gaussian illumination, one FFT to the pupil, rotation into the lab frame, then convolution along dispersion with the 7.7 nm pulse spectrum. The model is unfitted, yet reproduces two independent handoff constants — the 7.2 mm cross-dispersion footprint and the 0.3604 patch-edge intensity. No pattern clips the aperture (worst case 0.01% of power outside); the beam uses only ~41% of the aperture width across groove, which is structural, not adjustable. The 1.1514 anamorphic factor is omitted because the handoff fixes its magnitude but not its sign at the pupil; it would stretch the dispersion axis by ≤12% and moves no order across the aperture.

▸ source code